RAJALAKSHMI ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND DESIGN

DESIGN THINKING AND INNOVATION

REVIEW 1 REPORT

EMPATHIZE AND DEFINE

Academic Year	20____ - 20____	**Year / Semester**	_____
Section	_____	**Review Date**	____ / ____ / 20____

TEAM DETAILS

Team Name	PixelMinds
Project Title	MamaMind
Technology & Domain	[ ] ML [ ] DL [ ] AI [ ] VR [ ] AR [ ] Animation [ ] Hybrid / Other
Application Area / SDG	Maternal Health & Well-being (SDG – Good Health and Well-being)
Faculty Coordinator	Dr. P. Revathy
Team Members	1. Roll No: 241701016 Name: Gokul Krishna IS 2. Roll No: 241701057 Name: Sriyaanchita S
Team Contact / Email	

REVIEW OBJECTIVE

This review evaluates the team's ability to understand real users and stakeholders, collect authentic evidence, synthesize meaningful insights, and frame a focused human-centred problem. The same process applies to projects using ML, DL, AI, VR, AR, Animation or a hybrid technology; the technology should emerge as a suitable response to a validated need rather than as a predetermined solution.

PROJECT CONTEXT AND INITIAL SCOPE

Problem Context	*Where, when and for whom does the problem occur?* The postpartum period is a major life transition where new mothers and their partners experience physical, emotional, and lifestyle changes. Many parents struggle to balance baby care with their own well-being, often lacking personalized guidance, emotional support, and effective communication during this phase.
Primary Users / Beneficiaries	*Identify the main user group(s), not only the client or sponsor.* New mothers (primary users) Partners
Key Stakeholders	*Users, caregivers, teachers, administrators, experts, service providers, decision-makers, etc.* New mothers, partners/fathers, hospitals, obstetricians & gynecologists, and pediatricians involved in postpartum and newborn care
Initial Assumption to Validate	*State the team's starting assumption without presenting a final solution.* New parents need a single platform that combines baby care management, personalized wellness support, and shared parenting tools to help them navigate the postpartum journey with greater confidence and support.

USER RESEARCH SUMMARY

Research Methods Used	[ ] Interview [ ] Observation [ ] Survey [ ] Contextual Inquiry [ ] Expert Consultation
Number of Users / Stakeholders	10 New Parents (Survey)
Types of Users / Stakeholders	New Mothers
Research Locations / Platforms	Google Forms Interviews
Total Duration of Field Study	12 days

Sampling / Selection Rationale	Participants were selected because they have experienced or are currently experiencing the postpartum period and can provide first-hand insights into parenting challenges, emotional well-being, and baby care needs.
Important Limitations	• Limited sample size • Time constraints • Privacy concerns regarding personal experiences • Responses may vary based on individual circumstances • Limited access to healthcare professionals

EVIDENCE CHECKLIST

Evidence Type	Submitted (Yes/No)	File / Annexure Ref.	Faculty Verification / Notes
Research plan and interview / observation guide			
Interview notes, transcripts, survey responses or observation records	YES	Survey and empathy map	
Photographs, screenshots, sketches or field artefacts (where permitted)			
Persona / user profile and empathy map	YES	Empathy map	
Existing-solution / competitor / workflow study	YES	Studied BabyCenter, Huckleberry, and Bump apps to identify their features, differences, and unmet user needs.	
References for secondary research and domain facts			

EXISTING SOLUTION / CURRENT WORKFLOW STUDY

Existing Solution / Practice	What Works Well	Gaps / Unmet Needs
BabyCenter	Baby milestone tracking, parenting articles, baby care guidance	Limited focus on both parents' well-being and shared parenting.
The Bump	Pregnancy and baby tracking, educational resources	Lacks personalized emotional support and partner collaboration features.
Huckleberry	Feeding and sleep tracking, reminders	Primarily focuses on baby care; limited support for parental wellness and communication.

RESEARCH INTEGRITY AND INCLUSION CHECK

Privacy and Consent	*How were identities, photos, recordings and personal data handled responsibly?* Survey responses were collected voluntarily with participant consent. Personal identities were kept confidential, and the collected data will be used only for academic purposes.
Accessibility and Inclusion	*Were diverse abilities, languages, devices, connectivity and socio-economic contexts considered?* The survey was designed using simple language and made accessible online through Google Forms so that both mothers and partners could participate using smartphones or computers.
Bias and Fairness	*What sampling, data or representation biases may affect the findings?* The findings are based on responses from a limited number of participants, which may not represent the experiences of all new parents. Individual parenting experiences may vary depending on personal, cultural, and socio-economic factors.
Technology Feasibility Signals	*Only preliminary constraints: data access, hardware, immersion comfort, computational resources, content pipeline, etc.* The proposed application should provide an intuitive user interface, support secure data storage, enable AI-powered conversational guidance, and work efficiently on common Android smartphones with reliable internet connectivity.

INSIGHTS EXTRACTED

KEY PAIN POINTS / BARRIERS	KEY GAIN POINTS / EXPECTATIONS
• **New mothers often prioritize their baby's needs over their own well-being, leaving little time for self-care.** • **Partners are willing to help but often lack clarity on how to provide meaningful emotional and practical support.** • **Many new mothers experience feelings of isolation and have limited opportunities to connect with others going through similar experiences.** • **Parents often use multiple apps and resources for baby care, wellness, and parenting information, making it difficult to manage everything in one place.**	• **A unified platform that combines parent wellness, baby care, and parenting support in one place.** • **Simple tools to track feeding, sleep, vaccinations, medications, and baby growth milestones.** • **Positive motivation through small-win celebrations, daily reminders, and personalized encouragement.** • **Better communication and shared parenting by helping both parents understand and support each other.** • **Reliable, AI-powered guidance that provides quick answers to parenting and postpartum-related questions.** • **A safe and supportive environment where mothers can connect, share experiences, and learn from one another.**

INSIGHT STATEMENTS

Write 2–4 evidence-based, non-obvious insights. Each insight should connect an observed behaviour or pattern with an underlying need or motivation.

1	New mothers often prioritize their baby's needs over their own well-being, indicating the need for gentle reminders and positive encouragement to practice self-care.
2	Parents rely on multiple apps and information sources for baby care, indicating the need for a unified platform that combines wellness support, baby tracking, and parenting guidance.
3	New mothers often feel isolated during the postpartum period, showing the importance of creating a supportive community where they can connect with others experiencing similar journeys.
4	Shared parenting becomes more effective when both parents have visibility into baby-care responsibilities and can communicate their support needs with mutual consent.

PROBLEM FRAMING

POINT OF VIEW (POV) STATEMENT

[User / stakeholder] needs a way to [need] because [evidence-based insight].

New parents need a way to balance baby care with their own well-being because the demands of caring for a newborn often leave little time for self-care, effective communication, and accessing reliable postpartum support.

HOW MIGHT WE (HMW) STATEMENTS

1. How might we help new parents balance baby care while also making time for their own well-being?
2. How might we encourage both parents to actively participate in shared parenting responsibilities?
3. How might we help partners better understand and support each other during the postpartum journey?
4. How might we motivate new parents by celebrating their daily progress and small achievements instead of focusing on challenges?
5. How might we create a safe and supportive space where new mothers can connect, share experiences, and learn from one another?

REVIEW 1 SUBMISSION READINESS

The project problem is stated from the user / stakeholder perspective, not as a technology feature.	**[ ] Yes [ ] No**
Primary research evidence is organized and can be verified by the review panel.	**[ ] Yes [ ] No**
Insights are traceable to research observations, quotations, patterns or artefacts.	**[ ] Yes [ ] No**
The POV and HMW statements remain solution-neutral and allow multiple concepts.	**[ ] Yes [ ] No**
Every team member is prepared to explain their individual contribution during viva.	**[ ] Yes [ ] No**

Team declaration: The submitted field evidence is authentic, ethically collected and accurately represented. Any use of AI tools for transcription, summarisation, translation or visual preparation has been disclosed to the faculty coordinator.

GENERALIZED EVALUATION GUIDE – APPLICABLE TO ALL TECHNOLOGY DOMAINS

Criterion	Max	What the Faculty Evaluates
Research Depth & Authenticity	**5**	Relevant user groups and stakeholders; adequate and diverse primary research; genuine field engagement; context understood; claims not based only on assumptions.
Quality of Evidence Submitted	**3**	Organized, dated and traceable evidence; suitable notes, transcripts, survey/observation records, photos or artefacts; evidence directly supports findings; privacy respected.
Insight Clarity & Relevance	**4**	Findings are synthesized into clear pains, gains, needs and opportunity areas; insights are evidence-based, specific, non-obvious and relevant to the selected application context.
POV & HMW Quality	**3**	POV clearly identifies the user, need and insight. HMW statements are open-ended, actionable, solution-neutral, human-centred and neither too broad nor too narrow.
Understanding of User Research – Viva	**4**	Student can explain methods, users, evidence, findings, limitations, ethical issues and domain context, and can justify how conclusions were reached.
Clarity in Explaining Insights	**3**	Student communicates the research-to-insight-to-problem-framing logic clearly, using specific evidence and appropriate terminology.
Contribution & Involvement	**3**	Student shows verifiable ownership of assigned tasks, participation in field work and synthesis, collaboration, timely documentation and responsible conduct.

Scoring note: Award full marks only when the relevant indicators are demonstrated with convincing evidence. Award proportional marks when evidence is partial, weakly connected, generic or unsupported. Technical sophistication alone should not compensate for weak user research or problem framing in Review 1.

DOMAIN-NEUTRAL INTERPRETATION NOTES

ML / DL / AI	Evaluate the human need, decision context, data availability, representation, privacy, bias and explainability signals. Model accuracy or architecture is not the focus of Review 1.
VR / AR	Evaluate the user setting, device access, interaction needs, safety, motion comfort, spatial constraints, accessibility and the value of immersion or augmentation.
Animation	Evaluate the intended audience, communication or learning need, narrative comprehension, cultural relevance, accessibility, viewing context and media constraints.
Hybrid Projects	Apply the same human-centred criteria across the combined technologies. The team must justify why each technology is relevant to the validated need.
All Domains	Reward evidence-based problem understanding and clear framing. Do not reward a fashionable technology choice when the user need, evidence or opportunity is weak.

REVIEW 1 – MARK DISTRIBUTION (25 MARKS PER STUDENT)

Part A – Team Evaluation (Common Marks) – 15 Marks

Criteria	Max Marks	Marks Awarded	Brief Faculty Observation
Research Depth & Authenticity	5		
Quality of Evidence Submitted	3		
Insight Clarity & Relevance	4		
POV & HMW Quality	3		
Subtotal – Team Score	**15**		

Part B – Individual Evaluation – 10 Marks (Per Student)

Criteria	Max Marks	Student 1 Roll No: _____	Student 2 Roll No: _____	Student 3 Roll No: _____
Understanding of User Research (Viva)	4			
Clarity in Explaining Insights	3			
Contribution & Involvement	3			
Subtotal – Individual Score	**10**			

CONSOLIDATED MARKS

Roll No.	Student Name	Team Score /15	Individual Score /10	Total /25

COORDINATOR COMMENTS AND ACTION POINTS

Strengths / Evidence Accepted / Corrections Required / Next Review Expectations			
[ ] Approved	**[ ] Approved with Revision**	**[ ] Re-review Required**	**Re-review Date: ____ / ____ / 20____**

Signature : Faculty Coordinator (Dr. P. Revathy,Professor,CSD) **Head of the Department**